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PALEOMAGNETISM OF PRECAMBRIAN ROCKS FROM EL PASO, TEXAS

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Of six Precambrian rock types sampled at 12 sites in the Franklin Mountains, El Paso, Texas, a remanence which is stable to AF and/or thermal demagnetization has been isolated from Red Bluff granite, Franklin Mountain rhyolite, Fusselman Canyon diabase and hornfels in the Castner marble. Additional stability test are provided by reversals in the igneous rocks, the agreement of directions in the granite and its baked margin, a conglomerate test using granite cobbles from Recent Bolson deposits, and the randomness of directions in a tuffisite horizon at the base of the rhyolite pile. After structural correction the mean directions of magnetization from these four units are identical and give a paleomagnetic pole at 160° W, 28° N ($\delta p = 5°$, $\delta m = 6°$), which is considered to represent the field axis at the time of granite/rhyolite emplacement at 950×10^6 y. An older magnetic event may be recorded in the Castner hornfels. The 950×10^6 y pole is northeast of poles from North American rocks of the same age, but is identical with a pole from the 935×10^6 y Scottish Lower Torridonian Series, making allowance for continental drift: this suggest that a short period excursion of the field axis occurred during $935–950 \times 10^6$ y away from the main polar wandering path.

1. Introduction

The Franklin Mountains are a north trending range of mountains rising 3000 feet above a plain of unconsolidated Cenozoic deposits in far west Texas. They are typical of the Basin and Range province of the western United States, and consist of a block of Precambrian igneous and sedimentary rocks, and Phanerozoic sediments, tilted to the west. The igneous rocks are isotopically dated, so there is an excellent opportunity for a paleomagnetic study of rock types which are different in age but structurally and stratigraphically related.

2. Geology and age relations

HARBOUR (1960) summarized previous studies of Precambrian rocks in the El Paso area and his observations and the considerable detail of MCANULTY (1967) have added significantly to an understanding of the geology. An excellent summary of isotopic ages, with new data, and a regional synthesis is provided by DENISON and HETHERINGTON (1969).

The best exposures of the Precambrian section are in Fusselman Canyon (fig. 1). The Castner marble is the oldest unit in the Franklin Mountains and includes over 1000 feet of limestone, dolomite and hornfels, intruded by several thin diabase sills (these will be referred to as Fusselman Canyon diabase). The Mundy breccia overlies the Castner unit unconformably, and is up to 250 feet thick. Resting on this are 2600 feet of Llanoria quartzite, which is predominantly a massive, fine grained, cross-bedded, pure quartzite with subordinate argillaceous horizons. Overlying the Llanoria quartzite is a sequence of as much as 1400 feet of rhyolite belonging to the Thunderbird rhyolite (RICHARDSON, 1909), but more commonly termed the Franklin Mountain rhyolite (DENISON and HETHERINGTON, 1969). Most workers consider the rhyolite to be a porphyritic extrusive, and as far as structures will allow, it appears to be roughly conformable with the underlying Precambrian sediments.

Massive granite and microgranite sills intrude the entire Precambrian sequence with mild contact-metamorphic effects (mainly hornfelsing). NELSON (1940) named this the Red Bluff granite. Its several phases, and possibly the presence of Tertiary granite in the Organ Mountains to the north (DUNHAM, 1935) led to the belief that post-Carboniferous granites also existed in the Franklin Mountains. This idea has been examined with varying degrees of conviction, and several liberal interpretations have been put forward (see

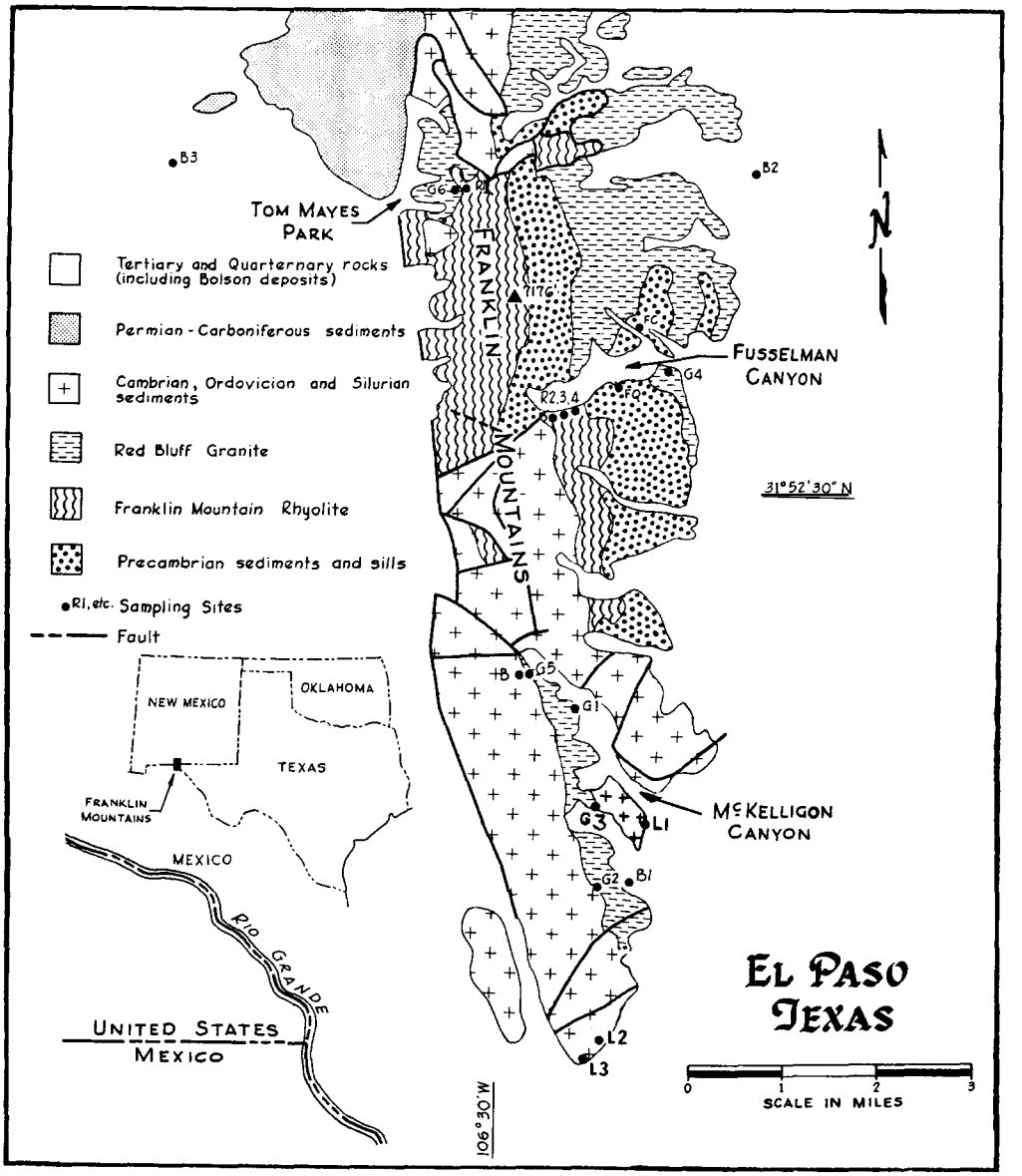


Fig. 1. Geologic map of Franklin Mountains, Texas (modified from Geologic Atlas: Van Horn – El Paso Sheet, Univ. Texas, Bur. Econ. Geol., 1967). For site details, see table 1.

LE MONE, 1969). What is obviously required is a detailed investigation of all the igneous rocks in the Franklin Mountains. The present study was limited to those phases which have been isotopically dated as Precambrian and their immediately associated, petrologically similar units.

DENISON and HETHERINGTON (1969) found no systematic difference between isotopic ages from the rhyolite and the different phases of the granite. They grouped together new Rb/Sr ages from whole rocks and feldspars to give an isochron of $953 \pm 13 \times 10^6$ y and concluded that Red Bluff granitic activity took place shortly after extrusion of the rhyolite. MUEHLBERGER *et al.* (1966) have pointed out similar, coeval Precambrian granite–rhyolite associations in the mid-continent region of the United States.

Lower Palaeozoic sediments overlie the Precambrian sequence with only slight disconformity (LE MONE,

TABLE 1

Sampling details. For site and location, see also fig. 1

Site	Location	Number of samples	Remarks	Attitude	
				Direction of dip, east of true north	Dip
1. Red Bluff granite. Block sampled					
G1	McKelligon Canyon	10	3000 ft below contact with (Cambro-Ordovician) Bliss sandstone	258	27
G2	Cable car tramway below Ranger Peak	7	Attitude assumed from McMillan Quarry limestone	226	34
G3	Head of McMillan Quarry	4	Attitude taken from Ordovician limestone	226	34
G4	Fusselman Canyon earth dam	4	Quartz syenite. Attitude taken from Precambrian sediments	233	21
G5	McKelligon Canyon	7	Just below erosion contact with Bliss sandstone	258	27
G6	Tom Mayes Park	4	Attitude assumed from adjacent Paleozoic rocks	230	35
2. Franklin Mountain rhyolite. Block sampled					
R1	Tom Mayes Park	11	1000 ft above granite contact. Attitude taken from adjacent Paleozoic rocks	230	35
R2	Fusselman Canyon	5	1100 ft below contact with Bliss sandstone. Attitude from this and indistinct banding in rhyolite	252	23
R3	Fusselman Canyon	4	1500 ft below contact with Bliss sandstone. Attitude from "flow" banding in rhyolite.	239	27
R4	Fusselman Canyon	3	Tuffisite near contact with Llanoria quartzite	[239]	[27]
3. Precambrian sediments and diabase in Fusselman Canyon					
A. Castner marble. Field drilled in HARBOUR (1960) section					
FC	150 ft above granite contact	10	White-green tremolite bearing marble	269	32
FC	200 ft above granite contact	14	Dark grey, calcareous hornfels with chert laminae	269	32
B. Fusselman Canyon diabase. Block sampled in Castner marble section of HARBOUR (1930)					
FC	Contact with granite	4	Sill 20 ft thick	281	32
FC	50 ft above granite contact	3	Sill 10 ft thick	273	31
FC	300 ft above granite	4	Sill 15 ft thick	280	32
C. Llanoria quartzite. Block sampled on south slopes of Fusselman Canyon					
FQ	1000 ft below rhyolite contact	4	Massive, bedded quartzite	262	19
FQ	1500 ft below rhyolite contact	4	Massive quartzite with shale lenses	258	24
FQ	2000 ft below rhyolite contact	4	Massive, bedded quartzite with slump structures	263	26

1969), implying a near-perfect peneplanation of the basement rocks, although LE MONE *et al.* (1967) describe hills of rhyolite which were remnants in the early Ordovician seas. Above these units rest massive, mid-Silurian dolomites and an incomplete Devonian to Cretaceous section. All the layered rocks in the Franklin Mountains dip westwards between 25 to 35°, confirming that the range is a tilted block. Several faults exist within the block itself (fig. 1); they are probably of pre-mid-Tertiary age and are discussed by MCANULTY (1967). In the areas visited, these have produced displacement without contortion and brecciation of the rocks.

Extensive deposits of Bolson extend for several miles on either side of the Franklin Mountains, and they offer the possibility of a conglomerate test (GRAHAM, 1949). Bolson consists of a heterogeneous mixture of sub-rounded clasts of granite, rhyolite and sedimentary rocks, firmly cemented by caliche. RIGHTMIRE (1967) determined radiocarbon ages on caliche from Hudspeth County, Texas, to the east and these varied from 1700 y B.P. to 'dead'.

3. Sampling details and experimental procedures

Three main collections were made from the rocks exposed in the Franklin Mountains: site details are shown in fig. 1 and table 1.

Both block sampling and field drilling methods were used. Orientation for the former was by both magnetic and sun compass, and for the latter by sun compass; the angular differences between the two orientation methods were a few degrees, when a declination of 12° E was assumed from the 1965.0 Isogonic Charts (USCGS). Structural correction for the granite and rhyolite were taken mainly from dips observed in the overlying sediments, whether Precambrian or Paleozoic. At most sites, the appropriate attitude is clear; at others it has been assumed, although the approximate parallelism between Precambrian and Phanerozoic layered rocks makes this assumption justifiable. In the rhyolite, indistinct (flow ?) banding is sometimes observed. In Fusselman Canyon, 24 samples were field drilled in hornfels and marble of the Castner section, and three diabase sills and three sites in the Llanoria quartzite were block sampled. Cobbles of the Red Bluff granite were collected at three sites in the Hueco and Le Mesa Bolson (B1, B2, B3 in fig. 1).

In the laboratory, cylinders of diameter and length 1 inch were cut from the sedimentary and diabasic rocks. Apart from site 1 samples, the granite and rhyolite proved difficult to drill, and were therefore cut into cubes of dimension 1.1 inches.

Progressive thermal and alternating field (AF) demagnetization each were applied to one specimen drilled from each sample. The equipment is located in a shielded mu-metal room (residual field for AF $<$ 200 γ; for thermal $<$ 10 γ) and is described by HELSLEY and SPALL (1970). NRM measurements were made on a PAR spinner magnetometer, model SM-1.

4. Red Bluff granite and Franklin Mountain rhyolite

These two rock types are discussed together because isotopically, their ages are equivalent, and thus their magnetization should also be related.

4.1. *Pilot demagnetization*

One specimen from each of the samples at each site was subjected to AF demagnetization in successive stages of 350, 700 and 1400 Oe. The NRM data were corrected for dip according to table 1, and site means were computed using the methods of FISHER (1953). Specimens from two sites in the granite and one in the rhyolite were thermally demagnetized to one or more of the temperatures 300, 400, 500 and 600 °C. The data were processed in the same manner as for AF demagnetization. By this means, k is obtained which is an estimate of the dispersion of the NRM vectors. The change in dispersion during demagnetization can be conveniently followed by the ratio k/k_0, where k_0 is the dispersion for a random population (see IRVING *et al.*, 1961). This method is useful for normalising the dispersion from different populations with unequal numbers of samples. In addition, the tabulations by MCELHINNY (1964) allow the dispersions after successive demagnetization stages to be compared to test whether they are significantly different.

At site 1 for both rock types, only a few of the samples were subjected to the pilot demagnetization experiment; this was due mainly to the larger number of samples collected at these sites. Demagnetization of the rhyolite at site 1 included an extra state of 175 Oe, but it was omitted from other sites because the change in NRM direction was very small.

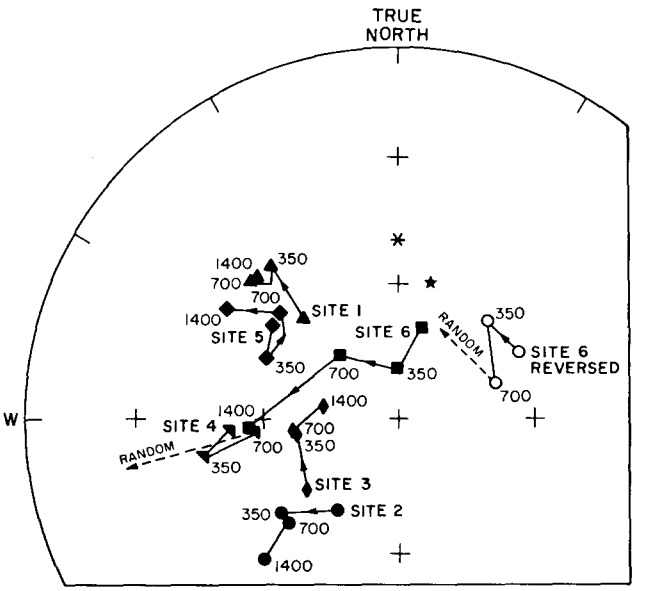


Fig. 2. Progressive AF demagnetization paths (structurally corrected) for each site in the Red Bluff granite plotted on an equal area projection. Solid symbols are downward seeking directions, open symbols are upward seeking directions. Fields are in Oe. Asterisk is the axial field, star is the inclined present field.

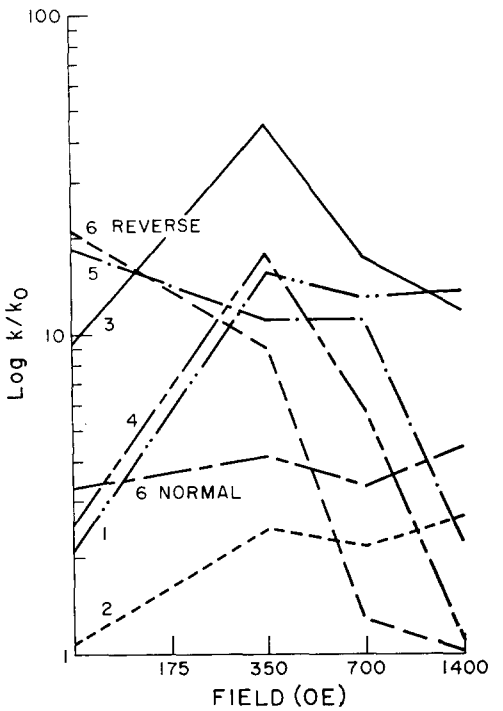


Fig. 3. Change of dispersion (measured by log k/k_0) for each site during AF demagnetization.

4.2 *Red Bluff granite*

The justification for omitting low field demagnetization is shown in fig. 2, because there is less than 15° of angular change between the initial NRM direction and that at 350 Oe for each site. In fact, there is about the same angular change between the 350 and 700 Oe stages, although it is not always along the same great circle, i.e. it is not systematic. This is reflected in the k/k_0 ratios at each stage (fig. 3) which show that the dispersion does not significantly change (at the 95% confidence level) between the 350 and 700 Oe stages for all the normal specimens.

That the mean vector change to 350 Oe is meaningful is demonstrated by a significant decrease in within-site dispersion of the normal directions at three of the six sites (1, 3, 4). Site 2 data are discussed below. At site 5, there is least dispersion among the *initial* NRM directions. However, it is not significantly different at the 95% level of confidence, from the dispersion at 350 and 700 Oe, and the initial mean lies midway between those at 350 and 700 Oe. It implies that the NRM is stable, and that AF demagnetization is merely producing random fluctuations in the remaining small secondary components. At site 6, using normal NRM directions, there is *systematic* vector rotation to 1400 Oe. Since there is only a slight (but not significant) decrease in the dispersion, then the overall direction change is meaningful. Furthermore it is *away* from the present field direction. (Note that for 4 and 5 at the 1400 Oe stage, the dispersion significantly *increases*, whereas for sites 1 and 3 there is little change.)

The overall effect of AF demagnetization is to remove small, in most cases, randomly oriented magnetizations by treatment in 350 Oe. In higher fields, the dispersion increases, although the direction does not markedly change. For only one site, there is systematic vector rotation to 1400 Oe. It is therefore a reasonable conclusion that blanket demagnetization to 1400 Oe for site 6, and to 350 Oe for the other sites, will effectively remove secondary magnetizations.

For site 2 specimens, the within-site dispersion is low at all stages (fig. 3). In fact it is never more than a factor of 2 or 3 from being statistically random, and the dispersion between any two stages is not significantly different at the 95% confidence level. Inspection of the individual specimen directions shows that two populations are present, roughly 90° apart. It seems probable that this site has been subjected to some style of faulting or folding which apparently is not visible in the field. The site means are not very different from those at other sites, but it is felt that this is a coincidence. The site was therefore rejected from further analysis.

Reversed directions of NRM are present at two sites (1 and 6). They were confined to single specimens rather than whole samples, suggesting that the feature has a mineralogical rather than geomagnetic origin. Only at site 6 were enough present (3) to enable a separate mean to be computed (fig. 2). Demagnetization of the site 6 reversed specimens produced a marked *increase* in dispersion, so that it is almost random at 700 Oe. Recall that vector rotation continued up to 1400 Oe for the normal specimens from this site, so that the supposed self-reversal process may be linked with the acquisition of large secondary magnetizations. The site 1 reversed specimen changes less than 5° on AF demagnetization: it was not included in computations of the site 1 mean.

None of the reversed directions is exactly antiparallel to the normal directions. The initial mean of the reversed set at site 6 is 168° away from the normal site

mean at 1400 Oe. The single specimen at site 1 is 163° away from the site 1 mean, both these directions being at 350 Oe. If *no* structural correction is applied to the NRM directions, the axial separation *increases* by 20°, which suggests that the reversal is a pre-tilting condition. The fact that approximately opposite polarities do exist and that they are removed from the present field is further evidence that the magnetization is (i) stable and (ii) reliable. That they are not exactly antiparallel may reflect that a component of fixed direction and due to external sources (e.g. a local anomaly or a non-dipole component) has been added equally to both normal and reversed magnetizations. It may however be simpler to invoke an incomplete reversal process.

Thermal demagnetization supports the contention that the NRM is stable. For site 1, the overall vector rotation for three specimens to 600 °C is less than 3° (fig. 4), and the dispersion does not change significantly. The means during thermal demagnetization are within 5° of those found during AF demagnetization. The thermal decay curves suggest a thermally discrete component (IRVING and OPDYKE, 1965) with a single blocking (Curie ?) temperature near 600 °C. Since the site 1 experiment was made shortly after the furnace had been built and prior to thermal calibration, it is felt that the remanence measured at 600 °C reflects uncertainty in the determination of temperature rather than a mineralogical reason. Again, one specimen from a sample at site 1 is reversed and it is 158° away from the normal directions.

Specimens from site 5 were heated once, to 500 °C (fig. 4). The resulting mean is about 14° away from the AF site mean at 350 Oe. However, the dispersions from both methods are similar, and an F test at the 95% level (WATSON and IRVING, 1957) shows that there is no significant difference between the two directions. Further temperature stages would perhaps have improved the agreement in directions. Note that the initial NRM directions for both sets of specimens are markedly different; this is caused by one specimen in the thermally demagnetized set having a shallow upward inclination. It supports the idea that the reversed directions may be related to the acquisition of secondary magnetizations.

As a further test of stability, the NRM directions of six specimens were remeasured three years after they had been stored in the laboratory in positions roughly antiparallel to their field orientations. The average direction change was 8° (range 1 to 16°) which is considered to be insignificant because different instruments were used for the two sets of measurements.

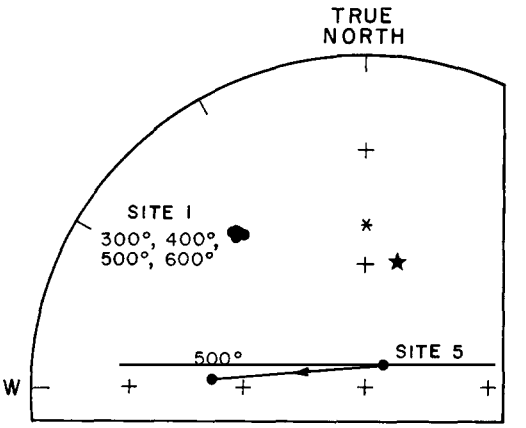


Fig. 4. Response to thermal demagnetization for granite specimens from sites 1 and 5 plotted on an equal area projection. Temperatures in °C; all directions are downward seeking. Axial and present fields are also shown.

4.3. *Franklin Mountain rhyolite*

Rhyolite from three of the four sites shows a response to AF demagnetization very similar to the granite. At sites 1, 2 and 3, there is less than 10° of vector rotation as far as 700 Oe, accompanied by a decrease in dispersion (fig. 5). At 1400 Oe however, the dispersion increases and the mean direction changes considerably. The dispersion at site 2 indicates a random distribution until the 700 Oe stage, although the mean directions are close to those from other sites. This is due to some specimens possessing shallow initial NRM directions. The result from this site is considered to be meaningful because the decrease in dispersion at 700 Oe is significant at the 95% confidence level. Five specimens from site 1 which were thermally demagnetized are also shown in fig. 5. The response is indistinguishable from that to the AF method, both in vector rotation and in final dispersion. A similar comment about uncertainty in temperature measurement applies to the directions at 600 °C as did for the granite specimens from site 1.

One specimen from site 2 was thermally demagnetized. Apart from one rather erratic observation, its direction moved consistently towards the means ob-

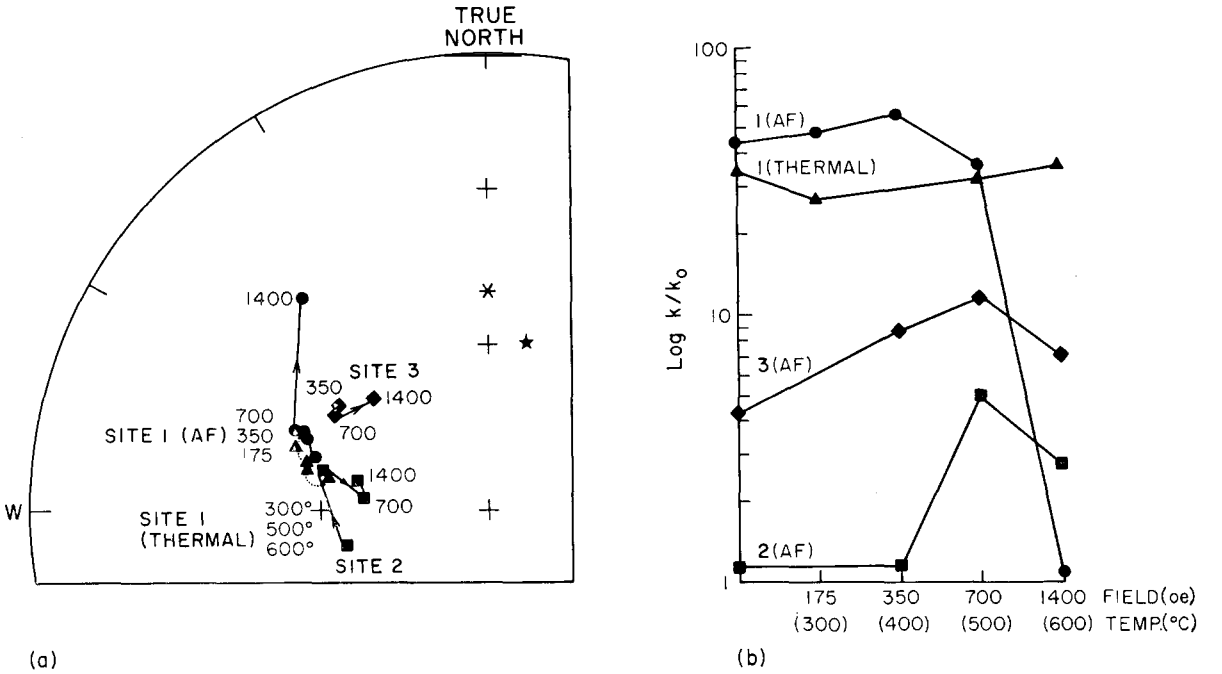


Fig. 5. (a). Thermal and AF demagnetization paths (structurally corrected) for specimens from sites in the Franklin Mountain rhyolite plotted on an equal area projection. All directions are downward seeking; triangles are thermal data. Fields in Oe, temperatures in °C. Axial and present fields are also shown. (b). Change in dispersion (measured by log k/k_0) for each temperature and field stage.

tained by AF demagnetization. The erratic change was puzzling and caused some concern, particularly since other rhyolite specimens show remarkable stability to both types of demagnetization. On checking the records of furnace heatings it seemed likely that a partial TRM had been induced in the rhyolite specimen during cooling by its proximity on the same shelf in the furnace to two specimens with moments two orders of magnitude higher.

As in the case of the Red Bluff granite, reversed directions occur (four in number), and with one exception they are confined to single specimens from different sites. This lends further weight to the argument for a self reversal process. The reversed specimens are stable to AF and thermal demagnetization. Using the normal site means at 700 Oe and 500 °C for comparison, the polarities differ by 167° for the specimen at site 1, and 110° and 169°, respectively, for the two specimens from site 2. The divergences *increase* when no structural correction is applied.

Seven specimens were stored for three years in positions opposite to their field orientations. They showed an average direction change of 11° (range 1 to 46°):

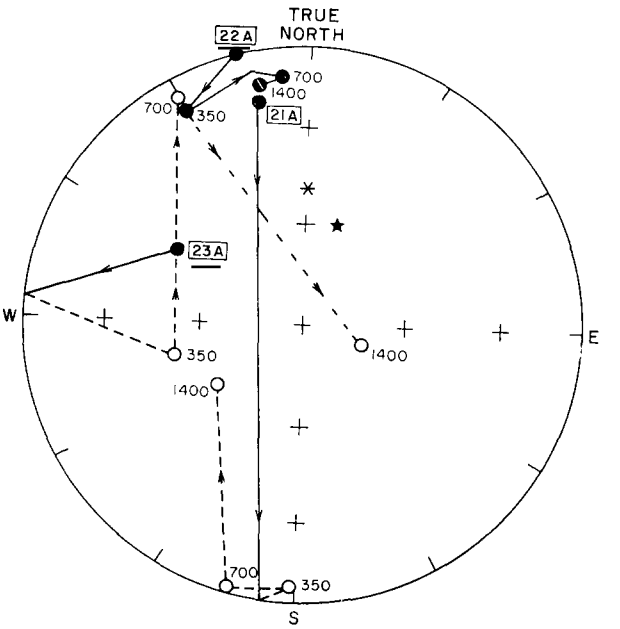


Fig. 6. Progressive AF demagnetization paths from 3 tuffisite samples (numbered) from site 4 plotted on an equal area projection. Solid symbols are downward seeking directions, open symbols are upward seeking directions. Fields in Oe. Axial and present fields are also shown.

the greatest change occurred in site 2 specimens, which have the poorest *initial* dispersion (fig. 5).

Site 4 is from a tuffisite horizon and its magnetization is anomalous. For two of the three specimens demagnetized (21A, 23A), there is an erratic vector movement of about 270° during AF demagnetization (fig. 6); the remaining specimen moves 60° but not in a systematic fashion (22A). The site represents a tuffisite breccia near the base of the rhyolite pile. Reaction rims around many of the larger fragments attest that they have been raised to a high temperature, perhaps during welding by succeeding rhyolite flows or by intrusion of the Red Bluff granite. The magnetic evidence suggests that the NRM is composed of a number of contributing magnetizations, all randomly oriented. Even if these are stable, as might be implied from the behaviour of rhyolite from other sites, a different fraction seems to be removed from each fragmentary magnetization after each stage, which is sufficient to ensure that the resultant vector remains random. Support for this comes from specimen 22A which includes a clast large enough to give a preferred direction to the NRM and thus only 60° of rotation on demagnetization. No further studies were made on samples from this site, and the data were not used in any of the summary calculations.

4.4. *Opaque mineralogy*

Two features about the opaque mineralogy are common to both granite and rhyolite. Firstly, there is an abundance of grains with dimensions less than 10 μm; secondly, titanomagnetite is oxidized directly to hematite. In addition, the granites are distinct in possessing some grains greater than 100 μm, and varying amounts of goethite.

The main Fe–Ti oxides in both rocks are cubic titanomagnetites which are directly oxidized to hematite along wedge shaped 111 planes and around the grain margins. The amount of oxidation varies within a specimen: it is never more than 75%. Along some grain margins, a blueish white colouration may signify titanomaghemite, but if so, it has no reflectivity contrast with the hematite. Scans with an electron microprobe kindly performed by J. L. Carter at the University of Texas at Dallas suggest that the titanomagnetite is essentially free of titanium.

The majority of the Fe–Ti oxide grains are less than 10 μm in dimension. Both rock types appear to contain the same volumes of Fe–Ti oxides. In the rhyolites, these are mainly submicroscopic material (at ×1250 magnification). In the granite they take the form of occasional large grains in the 100 to 200 μm range. Further distinctive features about the granites are:

(i) some pyrite in the quartz syenite from site 4;

(ii) the development of colloform goethite in vesicles and spongy hematite along cracks in the rock. The two minerals are not seen together.

There are no mineralogic differences between the normal and reverse specimens, except an *apparent* greater degree of oxidation to hematite in the reverse specimens. It would be difficult to make quantitative measurements of this oxidation. Firstly, hematite occurs in two modes. Secondly, the extent of oxidation varies within a specimen. Thirdly, much of the opaque material is submicroscopic.

4.5. *Curie points*

Four determinations were made on granite and rhyolite specimens in a field of about 2000 Oe on instruments at the University of Liverpool (built by R. L. Wilson) and at the University of Colorado (built by E. E. Larson). All the curves indicate sharp Curie points in the region 580 to 595 °C, with no indication of secondary blocking temperatures (fig. 7). Three of the curves are closely reversible. The difference in moment on cooling for the specimen in fig. 7a seems too great to explain as thermal lag, and is probably real. It may reflect changes in the mineral(s) responsible for the secondary magnetization. The interesting fact from these few determinations is that there is no evidence that hematite contributes to the saturation magnetization. Moreover, it supports the idea that a titanium-free magnetite is the main carrier of this remanence.

4.6. *Intensities, initial susceptibilities and Q′ ratios*

It was not possible to measure accurately the susceptibility of cubic specimens. Thus only a few data are available, mainly from site 1 in each rock type. Although there is a difference of about two orders of magnitude between the intensity and susceptibility ranges for granite and rhyolite (assuming arithmetic distributions), the *median* Q' ratios (intensity/susceptibility) are the same. (Only *median* values are quoted,

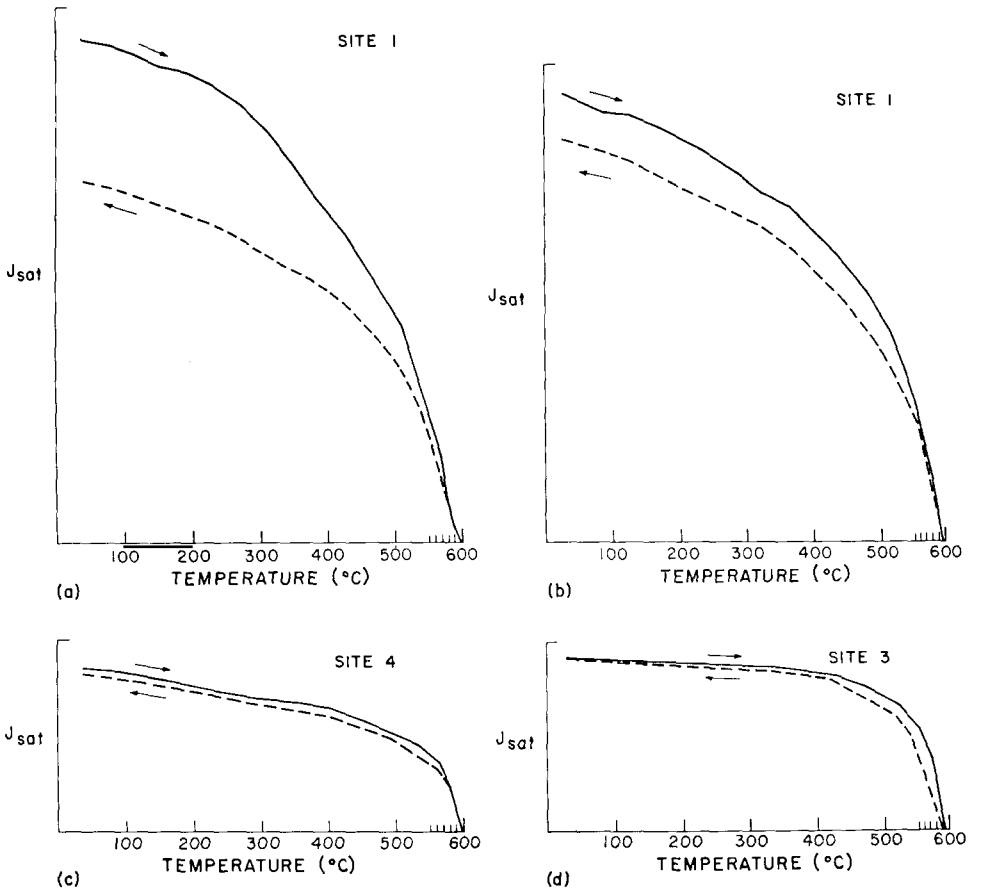


Fig. 7. Curie point determinations in a field of 2000 Oe on specimens of granite (*a*, *c*) and rhyolite (*b*, *d*). Instruments were at Liverpool (*a*, *b*) and Colorado (*c*, *d*).

firstly, because the number of specimens at each site is small, and secondly, to avoid skewing due to any lightning induced magnetization). Thus for site 1, the median Q' value for both rock types is 0.2. These are probably representative of the whole collection because susceptibility measurements from 20 specimens from other sites also give median Q' ratios of 0.2.

The range in intensity for *all* the granite specimens is from 1.0 to 143.0×10^{-6} emu/cm^3, with a median at 9.1×10^{-6} emu/cm^3. For the rhyolite collection, the intensity ranges from 0.2 to 23.3×10^{-4} emu/cm^3 with a median at 3.6×10^{-4}emu/cm^3. The intensity at site 4 (the tuffisite horizon) is an order of magnitude less than at other sites.

The susceptibility range for site 1 granite specimens is 32 to 46×10^{-6} emu/cm^3; for site 1 rhyolite, it is 26 to 65×10^{-4}emu/cm^3. Thus the suggestion from microscopic work that the *bulk* content of Fe–Ti oxide in each rock type is the same is opposed although it should also be noted that susceptibility increases with decreasing grain size (PARRY, 1965).

Interestingly, there are no marked differences in intensity or susceptibility between the normal and reversed specimens of either rock type. If anything, the latter possess slightly smaller intensities.

The very much higher intensity of the rhyolite does suggest that the remanence is carried by grains with dimensions much less than a few microns, as discussed by LARSON *et al.* (1969) so that the titanomagnetite approximates single domain behaviour (MORRISH and YU, 1955). In this respect, the coarser grain size in the granite is typical of remanence coercivities of about 350 Oe; the very fine grain size in the rhyolite has remanence coercivities of about 700 Oe.

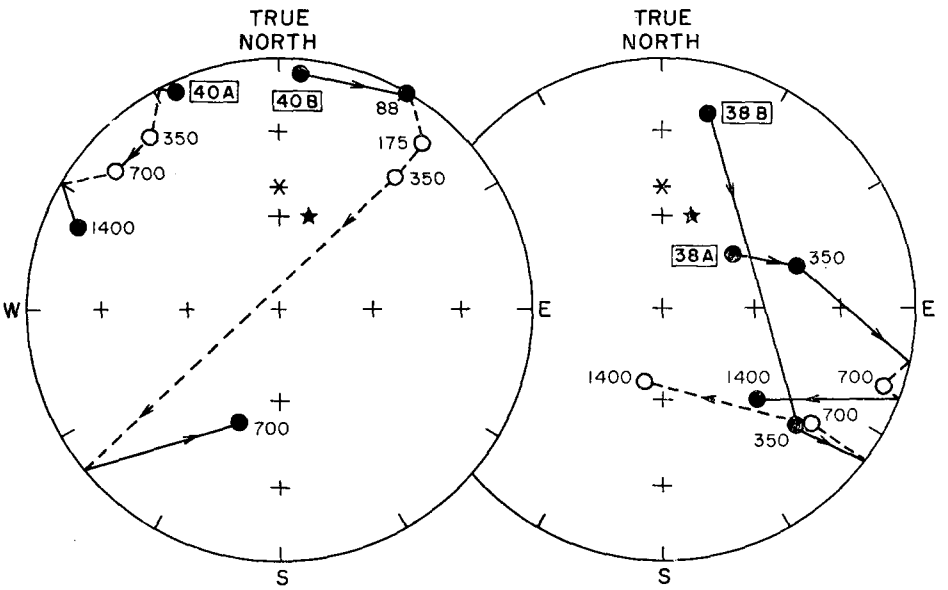


Fig. 8. Progressive AF demagnetization paths (not structurally corrected) for specimens from two unstable Bolson samples, plotted on an equal area projection. Solid symbols are downward seeking directions, open symbols are upward seeking directions. Fields in Oe. Axial and present fields are also shown.

4.7. *Artificial thermoremanence*

A thermoremanence was given to specimens with normal and reversed initial NRM by cooling them from 750 °C in a field of 0.3 Oe. In all cases, the TRM was "normal" and within 2° of the axis of the applied field. This suggests that the rocks are magnetically isotropic and further that the reversed magnetization is not reproducible in the laboratory. The TRM/NRM intensity ratio was an average of 25 for the granite and 8 for the rhyolite. It is difficult to account for the higher TRM/NRM ratio in the granite, particularly since the smaller grain size in the rhyolite was used to explain its higher NRM. It may indicate that many of the larger grains in the granite originally possessed a higher NRM but this has decayed either viscously or by subsequent alteration. The development of goethite in the granite supports the idea of alteration.

4.8. *Bolson deposits*

The NRM of five granite cobbles from Bolson deposits in the area was investigated by applying a conglomerate test. Specimens from four of these samples behaved similarly; two sets of AF demagnetization paths, with no structural correction, are shown in fig. 8. The behaviour of all four cobbles is quite different from that observed in the main granite. In particular, the moments are about an order of magnitude less than in the main granite, the NRM is not stable to AF demagnetization, there is little within-sample consistency and the directions even after structural correction would bear no resemblance to those from the main granite (fig. 2). The last observation *is* to be expected from a conglomerate test; however, the NRM is *not* stable to demagnetization. Only the 5th sample proved magnetically stable (fig. 9).

The hand samples are extensively leached, and show concentric weathering shells in which there are hematite concentrations. Opaque mineralogy reveals that little remains of the iron oxides common in the main

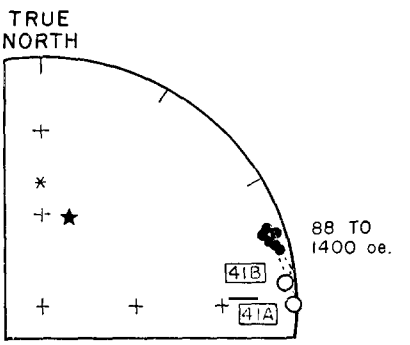


Fig. 9. Progressive AF demagnetization of specimens from a stable Bolson sample. Legend as in fig. 8.

TABLE 2

Mean directions of magnetization and paleomagnetic poles for granite and rhyolite. All data are structurally corrected according to table 1

Granite

Site nr.	Field (Oe)	Dec (°)	Inc (°)	N	R	k	α_{95} (°)	Long	Lat	δp	δm
1	0	320	57	10	9.267	12	14	175° W	57° N	15	20
1	350	327	49	10	9.788	43	7	168° E	61° N	6	9
3	0	222	67	8	7.861	51	8	131° W	1° S	11	13
3	350	248	69	8	7.865	52	7	141° W	13° N	11	13
4	0	265	62	8	7.506	14	15	155° W	18° N	19	24
4	350	256	61	8	7.475	14	15	153° W	11° N	19	24
5	0	303	61	14	13.572	30	7	166° W	44° N	8	11
5	350	288	61	14	13.270	18	9	163° W	32° N	11	15
6	0	9	69	8	7.640	19	13	91° W	68° N	18	21
6	1400	261	51	8	7.703	24	11	164° W	9° N	10	15
Mean	–	278	62	5	4.788	19	18	161° W	26° N	21	28

Rhyolite

Site nr.	Field (Oe)	Dec (°)	Inc (°)	N	R	k	α_{95} (°)	Long (° W)	Lat (° N)	δp	δm
1	0	277	60	11	10.734	38	7	162	25	8	11
1	350	273	58	11	10.466	19	10	162	21	12	16
2	0	279	60	10	7.761	4	27	159	19	31	41
2	700	281	68	10	9.593	22	10	152	31	15	17
3	0	300	61	8	7.698	23	12	166	42	13	18
3	700	305	58	8	7.861	51	8	171	45	8	11
Mean	–	287	62	3	2.970	67	15	162	32	18	23

Dec – declination of the north seeking magnetic vector, east of the true north
Inc – inclination of the north seeking magnetic vector, positive downward
R – length of N unit vectors, one specimen/site
$k = (N-1)/(N-R)$
α_{95} – semi-angle of the cone of 95% level confidence
Long – longitude of the paleomagnetic pole
Lat – latitude of the paleomagnetic pole
δp – semi-minor axis of the oval of 95% level confidence
δm – semi-major axis of the oval of 95% level confidence

granite. There is an abundance of long strings of spongy hematite and amorphous goethite throughout the entire rock. Goethite typically has a serrated outline, and often bifurcates in veins very like the pattern of a lightning stroke.

All this evidence suggests that the NRM of the cobbles is of chemical origin, perhaps acquired during rapid nucleation of hematite from solutions present during transport and deposition of the Bolson deposits. This accounts for their low moments, and for the lack of alignment of NRM directions along the present field, because the cobbles would be continually rolled about within the alluvial fans. The evidence from this conglomerate test is that while the condition for randomly oriented directions has been met, it is not conclusive proof of stability because the remanence of the cobbles has not always remained stable nor is there always within-sample consistency. In one sense, however, this is a stability test in itself. The samples from the main granite have also been subject to Recent weathering, albeit less severe, and certainly to prolonged Cambrian weathering at site 5, yet their NRM directions are well clustered and are removed from the present field. It is therefore argued that overall the evidence from the conglomerate test supports the idea that the magnetization of the granite is stable.

4.9. *Mean directions of magnetization and paleomagnetic poles*

The conclusion from the pilot demagnetization experiments is that the NRM of both granite and rhyolite contains only minor amounts of secondary magnetization. These are successfully removed by AF and thermal demagnetization. To give uniformity to the study, duplicate specimens from each sample were demagnetized to 350 or 1400 Oe for the granite, and to 350 or 700 Oe for the rhyolite, depending on which value produced minimum dispersion of the NRM directions. (For site 1, this involved one specimen from each of the remaining samples.) Three of these duplicate specimens were reversed initially. Since these are atypical of the whole suite, it was decided to demagnetize three more from the same samples, and these were normal. Site means were computed according to FISHER (1953) and also the corresponding paleomagnetic poles. They are listed in table 2 and shown in fig. 10 with the circles of 95% confidence.

In many cases, there was little change in the dispersion after demagnetization; for two sites, the dispersion actually worsened, although not significantly, at the 95% confidence level. Hartley's maximum F ratio test (WATSON and IRVING, 1957) shows there is no reason at the 95% confidence level to reject the hypothesis that the dispersion is not homogeneous over all the site populations in each rock type. The relevant equations are:

Granite: $k_{max}/k_{min} = 3.71$ and $4.10 < F_{8,17.2} < 5.19$
Rhyolite: $k_{max}/k_{min} = 2.68$ and $3.29 < F_{4,17.3} < 4.01$

By inspection, there is more dispersion among the granite means than among those from the rhyolite. In particular, site 1 is about 20° from the others; it is possible that this is also due to some kind of intra-site displacement as at site 2. However, since there were otherwise no differences between the sites, an overall mean has been calculated by combining sites 1, 3, 4, 5 and 6. The same was done for the three rhyolite means. They are also listed in table 2. An F test (WATSON and IRVING, 1957) shows that there is no significant difference between the two overall means, the relevant calculations being:

(i) Dispersion identity:
$k_1/k_2 = 3.52$ and $F_{8,4} = 6.04$;

(ii) Direction identity:
$F_{statistic} = 0.11$ and $F_{2,12} = 3.89$.

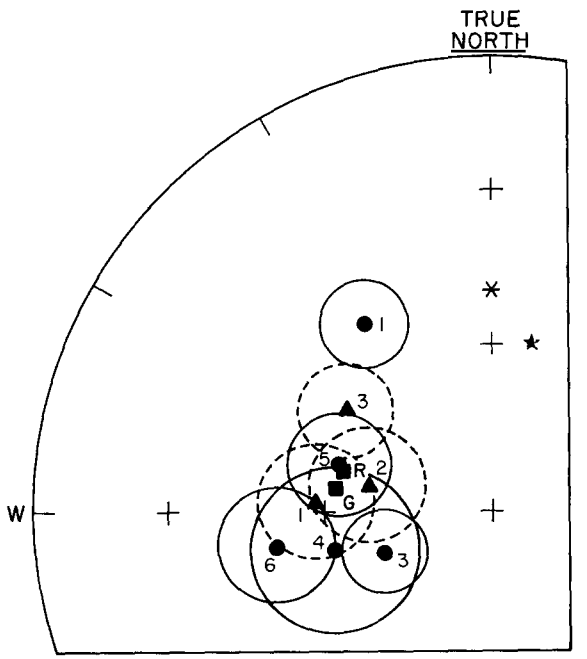


Fig. 10. Site mean directions of magnetization for granite and rhyolite, with associated circles of 95% level confidence, plotted on an equal area projection. All directions are downward seeking; circles are granite sites, triangles are rhyolite sites. Squares are the overall mean for granite (G) and rhyolite (R). Axial and present fields are also plotted.

5. Castner marble rocks

5.1. *NRM directions*

The NRM directions (structurally corrected) of 18 marble specimens (from 10 samples) and 24 hornfels specimens (from 14 samples) were combined according to FISHER (1953). The dispersion in the hornfels group is low ($\alpha_{95} = 6°$, $k = 25$) and all the directions are remote from the present field, in fact encompassing the means obtained for the granite and rhyolite. In contrast, the marble specimens possess a large overall dispersion ($\alpha_{95} = 25°, k = 2$), and there is considerable within-sample variation (the average angular change on remeasurement after eighteen months was 53°).

5.2. *Partial AF demagnetization*

Specimens from 11 of the hornfels group were subjected to AF demagnetization in 44, 175 and 350 Oe.

TABLE 3

Mean directions of magnetization during partial demagnetization of Castner hornfels. All data are structurally corrected. Legend as in table 2

Field (Oe)	Dec (°)	Inc (°)	N	R	k	α_{95} (°)	Long (° W)	Lat	δp	δm
0	273	60	11	10.589	24	9	161	22° N	10	14
44	266	61	11	10.654	29	8	158	17° N	10	13
175	253	59	11	10.518	21	10	154	8° N	11	15
350	244	58	11	10.433	18	11	151	2° N	12	16
500	235	30	3	2.685	6	54	162	19° S	32	59

The results are given in table 3 and fig. 11. A further stage at 500 Oe was intended, but only three specimens proved measurable afterward. The remainder were either too weak or unstable during measurement in the magnetometer. The result for these three specimens is also given but the dispersion is essentially random ($k/k_0 = 1.2$), so the direction is suspect. Such behaviour is incidentally surprising because hematite is the only recognizable magnetic mineral is polished section. Ten of the marble specimens were demagnetized to 175 Oe, but since there was no improvement in the dispersion before ($\alpha_{95} = 36°$, $k = 2$) and after ($\alpha_{95} = 39°$, $k = 2$) cleaning, no further studies were made on these samples. Their instability may be related to the fact that pyrrhotite is the only magnetic mineral observed in polished section.

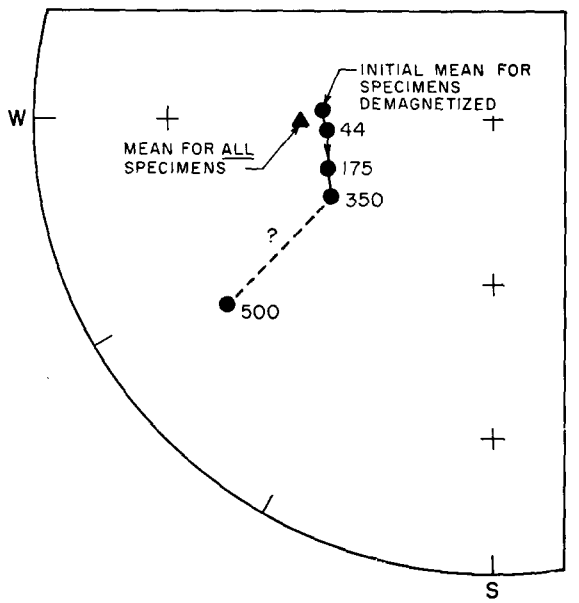


Fig. 11. Mean AF demagnetization path (structurally corrected) for 11 specimens of Castner hornfels, plotted on an equal area projection, together with mean for all hornfels specimens. All directions downward seeking. Fields in Oe.

The interesting result on AF demagnetization of the hornfels to 350 Oe, is that the dispersion does not change significantly at the 95% confidence level between each stage ($k = 24$, 29, 21 and 18, respectively). Yet there *is* a systematic change in direction of about 25°, which is shown by all the specimens. An F test (WATSON and IRVING, 1957) shows that the initial and final (350 Oe) directions are not significantly different at the 95% level, the relevant calculations being:

(i) Dispersion identity:
$k_1/k_2 = 1.37$ and $F_{20,20} = 2.12$;

(ii) Direction identity:
$F_{\text{statistic}} = 2.33$ and $F_{2,40} = 3.23$.

Nevertheless the argument can be put forward that the demagnetization path may in fact record the magnetic history of the rock. Since both structurally corrected *and* uncorrected initial mean NRM directions are removed from the present field, it is unlikely that the streaking is a Recent partial remagnetization. Two other mechanisms for the 25° of vector rotation can be suggested: secular variation and contact metamorphism.

It seems unlikely that secular variation has left a memory in the components of low coercive force because the streaking follows a great circle, not the circular pattern usually observed in secular variation.

The hornfels may have preserved a magnetic record of the heating accompanying intrusion of the granite at 950×10^6 y. It is unfortunate that, presumably because of the calcite content, no thermal demagnetization experiments were possible because the specimens exploded at about 300 °C. Nevertheless, the hypothesis

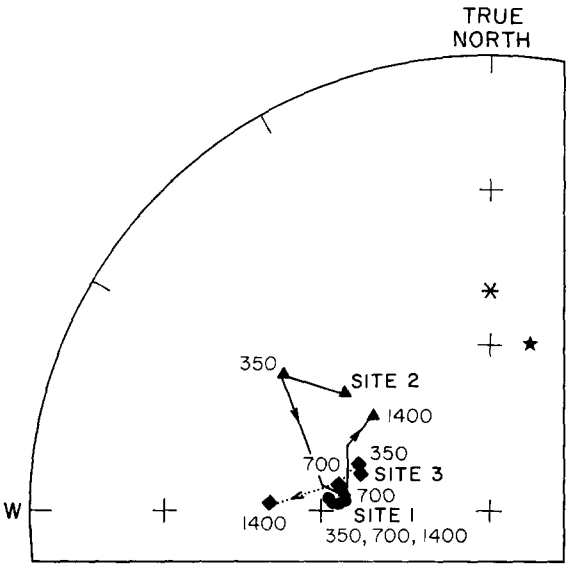


Fig. 12. Progressive AF demagnetization paths for the three Fusselman Canyon diabase sills (structurally corrected) plotted on an equal area projection, all directions being downward seeking. Fields in Oe. Axial and present fields are also shown.

can be tested further because the Fusselman Canyon diabase should also have preserved a memory of this contact metamorphism episode, and the initial NRM directions in the hornfels should accord with those in the granite. In this respect, an F rest on the initial mean from the hornfels and the overall granite mean shows that there is no significant difference between them, the relevant equations being:

(i) Dispersion identity:
$k_1/k_2 = 1.29$ and $F_{8,20} = 2.45$

(ii) Direction identity:
$F_{\text{statistic}} = 0.13$ and $F_{2,28} = 3.39$

6. Fusselman Canyon diabase

6.1. *AF demagnetization*

One specimen from each sample obtained from the three thin sills in the Castner section was demagnetized in AF stages of 350, 700 and 1400 Oe, and the site (sill) means computed using FISHER's (1953) method. At least one specimen from each site was subjected to thermal demagnetization in a field free space ($\pm$ 10 γ) in order to test whether a distinct change in the NRM directions occurred at a particular temperature.

The response to AF demagnetization is shown in fig. 12. No attempt is made to rigorously compare the dispersion in terms of k/k_0, because of the small numbers of specimens involved. In any case, the precision is universally high. There is no vector rotation for sill 1 even at 1400 Oe, and the dispersion remains constant (recall from table 1 that this sill is at the granite contact). For sill 2, there is about 40° of direction change in a roughly circular manner; the dispersion is significantly minimum at 700 Oe. For sill 3, there is less than 10° of vector rotation to 700 Oe and the dispersion remains low. It increases abruptly at 1400 Oe and there is an accompanying 15° direction change.

The conclusion from AF demagnetization is that 700 Oe is an effective cleaning stage, and that the sill means are then, by inspection, within a few degrees of each other. It is worth noting that the initial NRM mean for the Castner hornfels lies very close to the three diabase means at 700 Oe.

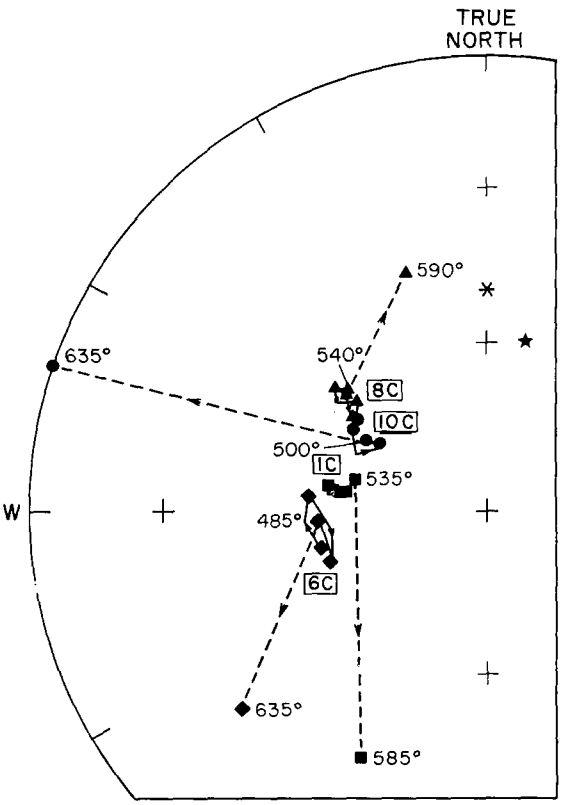


Fig. 13. Thermal demagnetization paths for four specimens from the Fusselman Canyon diabase, plotted on an equal area projection, with the final two temperature stages marked. All directions are downward seeking. Axial and present fields are also shown.

6.2. *Thermal demagnetization*

Thermal demagnetization up to 635 °C was applied to specimens from each sill. The individual response is shown in fig. 13, and by inspection the directions up to 540 °C do not markedly differ from the results obtained by AF demagnetization. There is less than 5° change in direction within the specimens until a temperature of 585 °C is reached, at which the change in direction is large and apparently at random. This is mirrored in the thermal decay curves which show an abrupt decrease in moment above 540 °C.

Two alternatives appear to be indicated by considering these observations alone. Firstly, the NRM of the diabase is a true and stable TRM due to a mineral with a Curie point between 540 and 585 °C, and there is no record of baking. Secondly, the diabase was hornfelsed by the granite to at least 540 °C, so that it acquired a partial TRM which completely obliterated the original TRM. It seems probable that both in fact contribute to the magnetic characteristics.

McANULTY (1967) argues that the Bowen series of mineral assemblages in the Fusselman Canyon rocks indicate a temperature of contact metamorphism between 450 and 500 °C. The magnetic evidence suggests about 540 °C, which is not unreasonable for the hornblende–hornfels facies (TURNER and VERHOOGEN, 1960, p. 520).

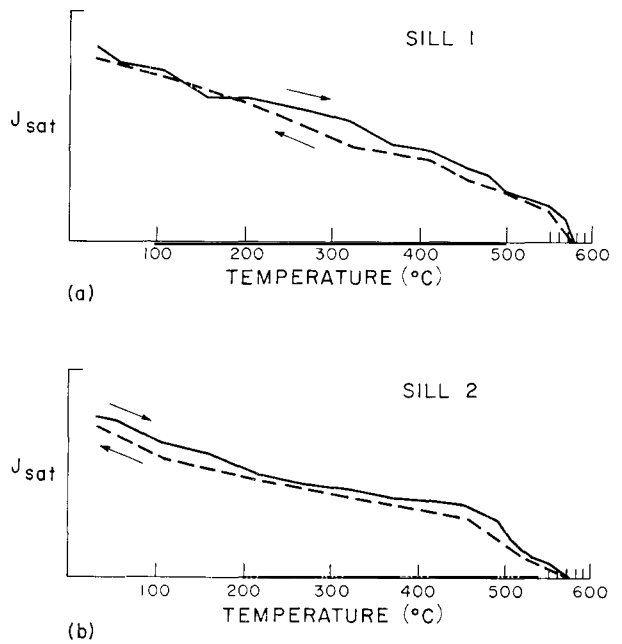


Fig. 14. Curie point determinations in a field of 2000 Oe on the Univ. of Colorado instrument.

6.3. *Opaque mineralogy*

Polished sections of the diabase at ×1250 magnification show that the opaque minerals are smaller than a few microns in size. Positive identification of all the phases is therefore difficult, but titanomagnetite, pyrrhotite and hematite are present. Secondary alteration is inferred from the abundant, rounded sphenes associated with the Fe–Ti oxides.

6.4. *Curie points*

Curie point determinations (fig. 14), do not support the idea that hematite contributes to the saturation magnetization. They rather indicate a titanomagnetite with a Curie temperature from 570 to 580 °C. Their low profile implies the presence of thermally distributed components, which is in direct contrast to the knee-shaped thermal decay curves. Note the break in slope at about 500 °C for the sill 2 specimen.

6.5 *Intensities, susceptibilities and Q′ ratios*

The intensity range of the three sills is from 4.9 to 72.8×10^{-5} emu/cm^3. It is slightly higher in the sill in contact with the granite. That this is due to remagnetization rather than metasomatism across the contact is suggested by the small overall range in susceptibility, which is from 7.6 to 12.9×10^{-5}emu/cm^3. The overall range in Q' ratio is from 0.6 to 6.1; the median value for each sill is 3.3, 1.3 and 1.1, respectively.

6.6 *Mean directions of magnetization and paleomagnetic poles*

Replicate specimens from each sample were AF demagnetized to 700 Oe because this stage produced minimum dispersion in the pilot experiments. Means, and the corresponding paleomagnetic pole positions, were computed for each sill; they are listed in table 4. All three means are identical, and furthermore they are very similar to both the initial NRM direction in the hornfels and the overall granite and rhyolite means. Identity tests are not possible because of the extremely high precision ($k > 999$) for the diabase result. It is considered that the stable magnetization in the Castner marble section records the field at the time of the 950×10^6 y igneous activity.

TABLE 4

Mean directions of magnetization and paleomagnetic poles for the Fusselman Canyon diabase. All data are structurally corrected. Legend as in table 2

Sill	Field (Oe)	Dec (°)	Inc (°)	N	R	k	α_{95}	Long (°W)	Lat (°N)	δp	δm
1	0	274	65	8	7.841	44	8	155	25	11	13
1	700	281	65	8	7.827	41	9	156	29	11	14
2	0	288	66	6	5.772	22	15	156	35	19	23
2	700	280	64	6	5.966	148	5	157	29	7	9
3	0	284	65	9	8.888	71	6	157	32	8	10
3	700	280	65	9	8.805	41	8	157	29	10	13
Mean	–	280	65	3	2.999	>999	1	157	20	1	1

7. Llanoria quartzite

It was hoped that the Llanoria Quartzite collection might offer a more positive indication of the pre-950×10^6 y field, especially as it came from a different structural block, presumably removed from immediate contact metamorphism effects. The response to thermal demagnetization was unfortunately disappointing and the results were considered to be of no paleomagnetic value. The main reason is that the NRM of the quartzite is both weak ($< 8 \times 10^{-7}$ emu/cm^3) and unstable during measurement.

8. Discussion

Of the six Precambrian rock types sampled in the Franklin Mountains, a remanence which is stable to AF and thermal demagnetization has been isolated from the Red Bluff granite, Franklin Mountain rhyolite, Fusselman Canyon diabase, and hornfels in the Castner marble. F tests at the 95% level of confidence indicate that the mean directions from the granite and rhyolite are not significantly different, and that the mean direction from the granite is not significantly different from the initial mean direction from the hornfels. The result from the diabase has too high a precision to be able to strictly conduct identity tests; however, its mean is consistent with the other directions. The fact that the NRM directions of the granite and its baked margin agree is a further test of stability.

The overall mean direction of magnetization obtained by combining the four rock means is at Dec 279°, Inc 62° ($\alpha_{95} = 4°$, $k = 557$), which gives an equivalent paleomagnetic pole at 160 °W, 28 °N ($\delta p = 5°, \delta m = 6°$). It is considered that this represents the axis of the field at the time of the igneous activity at 950×10^6 y. In the Castner hornfels, additional stable components are revealed by AF demagnetization. These may be an indication of the field at the time of deposition of the Castner sediments. The geologic age of this section is uncertain. DENISON (per-

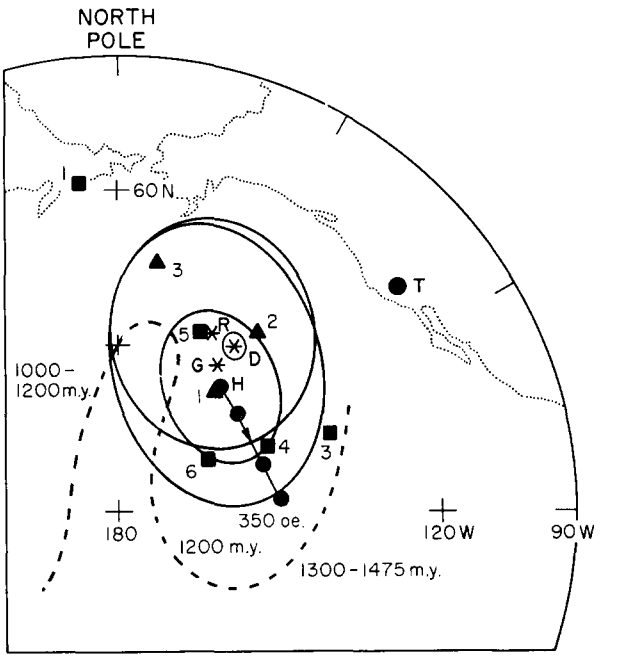


Fig. 15. Site mean paleomagnetic poles for the granite (squares) rhyolite (triangles) and the AF demagnetization path for the hornfels (H). Asterisks are overall mean poles for the granite (G), rhyolite (R) and diabase (D), with their ovals of 95% level confidence. T is a pole for the Lower Torridonian Series of Scotland (IRVING and RUNCORN, 1957). Apparent Precambrian polar wandering curve (with ages in 10^6 y) proposed for North America by SPALL (1970[a]) is plotted as a dashed line.

sonal communication) notes that similar sequences to the Llanoria quartzite are found in the De Baca subsurface terrane of New Mexico, and the Hazel Formation of the Van Horn area, Texas. The De Baca rocks contain rhyolite fragments which may have come from the Texas Panhandle subsurface volcanic terrane, dated at 1150 to 1200×10^6 y. Thus an older limit for deposition of the Castner section is possibly set at 1150 to 1200×10^6 y.

Paleomagnetic poles for the four stable rock types are given in fig. 15. If the hornfels does record a period of older field history, then it suggests that between 1150 and 950×10^6 y the paleomagnetic pole moved northward from the Pacific equator. It implies that the magnetic components of highest coercivity in the hornfels were unaffected by the igneous episode, and that this produced a circle of remagnetization which has remained unaffected by subsequent tectonic events in the area. It also suggests that the diabase sills have been *completely* remagnetized by intrusion of the granite.

There is a slight suggestion of a great circle path through all the means in fig. 15. If continued, this would pass close to Tertiary poles for North America (e.g. HOSPERS and VAN ANDEL, 1968). The argument for Tertiary remagnetization (during Basin and Range deformation) might thus be implied. This can be discounted because there is very little dispersion among the four *overall* means, as well as no streaking among the individual directions at each site, nor difference in response to AF and thermal demagnetization. Furthermore, this great circle trend disappears when no structural correction is applied, that is, such a remagnetization would have to be *before* tilting, and thus pre-Tertiary.

The poles recording the 950×10^6 y igneous activity and the ?1150 to 1200×10^6 y event recorded in the hornfels, are about 30° northeast of poles from Keweenawan rocks of similar age from North America (DU BOIS, 1962). In addition, the apparent polar wandering northward from the equator represented in these Texas rocks is opposite to the southward trend observed in the Keweenawan rocks (fig. 15 and DU BOIS, 1962; SPALL, 1970a).

The difference could be partially explained by horizontal displacement of the sampling area of the order of 2000 miles northeastward relative to the Keweenawan rocks of the Lake Superior region since 950×10^6 y. With the available geologic data this seems unlikely, although this argument cannot be entirely refuted until a better knowledge of the distribution of North American crustal plates during the Precambrian has been obtained.

Alternatively, both divergences from published paleomagnetic pole data might be due to the younger age of the Texas igneous activity and to the uncertain age of the "older" magnetic event recorded in the hornfels. In this respect, a comparison with poles obtained for dated 1300 to 1475×10^6 y igneous rocks from North America (fig. 15 and PALMER, 1969; SPALL, 1970a) suggests the Castner section may be older than 1150 to 1200×10^6 y. HOWELL *et al.* (1958) quote a pole for the Precambrian Hazel Formation in west Texas (of possibly similar age), which is 60° northwest of the "older" event in the Castner hornfels. However these red beds were not demagnetized, and IRVING (1964, p. 317) considers their pole to be unreliable.

A more promising explanation of the Texas 950×10^6 y poles is obtained by comparing them with one obtained for the Lower Torridonian Series from Northwest Scotland (IRVING and RUNCORN, 1957) which is isotopically dated as 935×10^6 y (MOORBATH, 1969). The two sets of poles are 40° apart (fig. 15), yet they become almost identical if Europe and North America are rotated to conform to the computer fit of BULLARD *et al.* (1964) and to the fit of selected paleomagnetic data (HOSPERS and VAN ANDEL, 1968). Making the assumption that the two continents *were* contiguous at that time, and that they have remained together until their break up, it suggests that the period $935–950 \times 10^6$ y was marked by a short period excursion of the Earth's field axis northeastward from the main polar wandering path observed in other rocks of similar age (DUBOIS, 1962; SPALL, 1970a).

9. Lower Paleozoic rocks

A continuous stratigraphic section of 55 samples (99 specimens) was drilled in the Cambro-Ordovician Bliss Sandstone in McKelligan Canyon (B in fig. 1), and 19 samples (54 specimens) were obtained from 3 sites in Ordovician limestone units on the eastern slopes of the Franklin Mountains (L1, L2, L3 in fig. 1). Exhaustive AF and thermal demagnetization studies failed to isolate any consistent preferred direction of magnetization from these rocks (SPALL, 1970b).

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